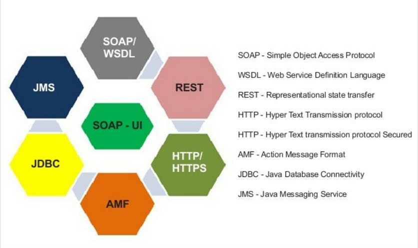

Figure 4: Protocols supported by Soap (Image credits: *guru99.com*)

Sahi

Sahi is an automation and testing tool for Web apps, available in an open source and proprietary version. The open source version includes all the tools required for most testing purposes, and is popular because it can be modified according to one's needs. Sahi Open Source is written using Java and JavaScript and has been hosted on Source Forge since October 2005. It has been released under an Apache License 2.0 and its latest version is 6.1 (published on November 5, 2014).

Sahi basically runs as a proxy server and the browser's proxy settings are configured so that they point to Sahi's proxy. Sahi then injects JavaScript event handlers into the Web pages, which allows it to record and then play back different events on the browser. Since Sahi uses the proxy to interact with different Web apps, it does not get affected by the browser on which that application is run.

Supported scripting language: Sahi Script
Features:

1. Can test Web applications running on IE, Chrome, Firefox, Safari, Opera and any modern browser.
2. The record and playback feature is available on all the browsers that it works on.
3. Generates output reports in HTML format whenever the script written is played back in order to test any action.
4. Can run a single test case or even suites of more than one test case at a time. It supports the batch-run process as well.
5. One can even play back more than one test script in parallel, at the same time, using Saki.

Apache JMeter

The Apache JMeter is open source software designed as a 100 per cent Java app to test functional behaviour and measure performance. It was originally designed for testing Web apps alone but has since expanded to other test functions as well. Apache JMeter can be used to test performance, both on static and dynamic resources, including dynamic Web applications.

It can also be used to simulate a heavy load on a single server, group of servers, network or even object—to test strength and to analyse the overall performance under different types of load.

Supported scripting languages: Groovy and Bean shell script
Features:

1. Can help in performing load testing as well as performance testing of the following different applications, servers and protocols:
   a. Web – HTTP and HTTPS (Java, PHP, NodeJS, SPINET, etc)
   b. Web services – SOAP/REST
   c. FTP and TCP protocols
   d. Database via JDBC
   e. Message-oriented middle ware (MOM) via JMS
   f. Native commands or shell scripts
   g. Mail – SMTP(S), POP3(S) and IMAP(S)
   h. LDAP and Java Objects
2. Consists of Test IDE which allows fast test plan building, recording (from browsers or native applications) and debugging.
3. Can run in command-line mode (non GUI/headless mode) to load tests from any Java compatible OS (Linux, Windows, etc).
4. Easily portable and generates a complete and dynamic HTML report.
5. Its multi-threading framework allows concurrent sampling by many threads, and even simultaneous sampling of different functions using separate thread groups.
6. Helps in caching and offline analysis or in replaying test results. END

References

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